Meeting Record

Mid point circle Drawing Algo:

(I) find the initial value of point $(0, r)$

(II) Calculate next coordinate either (x_k+1, y_k-1) or (x_k+1, y_k)

(III) Calculate mid point $(x_k+1, y_k-\frac{1}{2})$

(IV) Substitute the mid point in eqⁿ of circle or P_k

(V) Calculate P_k

(VI) Calculate P_{k+1}

(VII) Calculate $P_{k+1} - P_k$

(IX) Identify P_{k+1}

(X) Repeat the Condition $\boxed{x \geq y}$

if $P_k \geq 0$	if $P_k < 0$
$x_{k+1} = x_k+1$	$x_{k+1} = x_k+1$
$y_{k+1} = y_k-1$	$y_{k+1} = y_k$

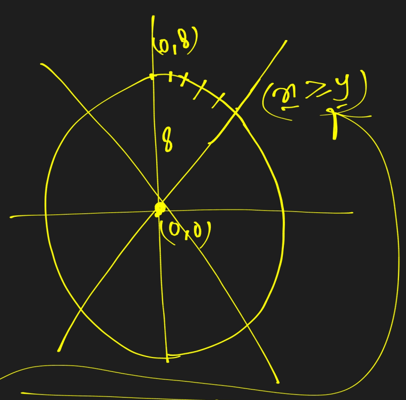
Q: Draw a circle with radius 8 using mid point Algo.

Solⁿ: $\boxed{P_k = 1-r}$

$P_{k+1} = P_k + 2(x_k+1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$

$x = 0,\ y = 8,\ r = 8$

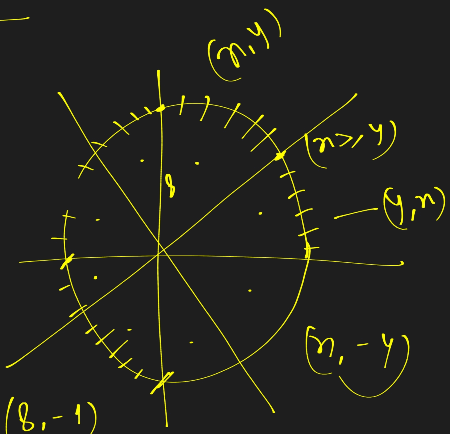
Step	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1	0	8	$P_k = 1-r = 1-8 = -7$	1	8
2	1	8	$P_{k+1} = -7 + 2(0+1) + (8^2-8^2) - (8-8)+1 = -4$	2	8
3	2	8	$P_{k+1} = -4 + 2(1+1) + 1 = 1$	3	7
4	3	7	$P_{k+1} = 1 + 2(2+1) + (7^2-8^2) - (7-8)+1 = -6$	4	7
5	4	7	$P_{k+1} = 3$	5	6
6	5	6	$P_{k+1} = 2$	$\boxed{6}$	$\boxed{5}$



if $P_k \geq 0$	if $P_k < 0$
$x_{k+1} = x_k+1$	$x_{k+1} = x_k+1$
$y_{k+1} = y_k-1$	$y_{k+1} = y_k$

(x,y) Q_1	$(-x,y)$ Q_2	$(-x,-y)$ Q_3	$(x,-y)$ Q_4
(0,8)	(0,8)	(0,-8)	(0,-8)
(1,8)	(-1,8)	(-1,-8)	(1,-8)
(2,8)	(-2,8)	(-2,-8)	(2,-8)
(3,7)	(-3,7)	(-3,-7)	(3,-7)
(4,7)	(-4,7)	(-4,-7)	(4,-7)
(5,6)	(-5,6)	(-5,-6)	(5,-6)
(6,5)	(-6,5)	(-6,-5)	(6,-5)

Q_1		Q_2		Q_3		Q_4	
(5,6)	(8,2)	(-5,6)		(-5,-6)		(5,-6)	
(6,5)	(8,1)	(-6,5)	(-8,2)	(-6,-5)	(-8,-1)	(6,-5)	(8,-1)
(7,4)	(8,0)	(-7,4)	(-8,1)	(-7,-4)	(-8,0)	(7,-4)	(8,0)
(7,3)		(-7,3)	(-8,0)	(-7,3)		(7,-3)	
				(-8,-2)		(8,-2)	



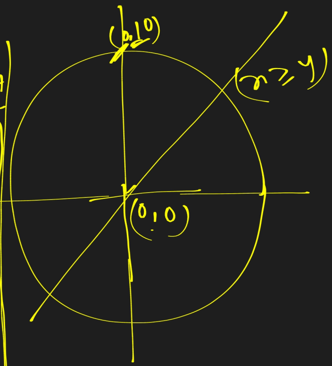
Q: Digitize the first octant of a circle by using Mid point Circle drawing Algorithm center at (10,20) and radius 10.

Sol^n :

$P_k = 1 - r = 1 - 10 = -9$

$P_{k+1} = P_k + 2(x_k+1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$

Steps	x_k	y_k	P_k	x_{k+1}	y_{k+1}	x_{plot}	y_{plot}
1	0	10	$P_k = -9$; $P_{k+1} = -9 + 2(0+1) + (10^2 - 10^2) - (10-10) + 1 = -6$	1	10	11	30
2	1	10	$P_{k+1} = -6 + 2\times2 + 1 = -1$	2	10	12	30
3	2	10	$P_{k+1} = -1 + 2\times3 + 1 = 6$	3	10	13	30
4	3	10	$P_{k+1} = 6 + 2\times4 + (9^2 - 10^2) - (9-10) + 1 = 6 + 8 - 19 + 2 = -3$	4	9	14	29
5	4	9	$P_{k+1} = -3 + 2\times5 + 1 = 8$	5	9	15	29
6	5	9	$P_{k+1} = 8 + 2\times6 + (8^2 - 9^2) - (8-9) + 1 = 5$	6	8	16	28
7	6	8		7	7	17	27



if $P_k \geq 0$

$x_{k+1} = x_k + 1$

$y_{k+1} = y_k - 1$

if $P_k < 0$

$x_{k+1} = x_k + 1$

$y_{k+1} = y_k$